Tutorial 1
Discussion: 20. October 2026, 22. October 2026

Task 1: Alternating current

Explain how alternating current and voltage can be modeled mathematically.

a) Describe them as a function of time.

b) Explain the following terms: **amplitude value**, **period**, **frequency**.

c) Graphically represent AC voltage or current.

d) How can the Root Mean Square (RMS) voltage value be calculated?

e) What is the peak (amplitude) value of the voltage measured from standard residential outlets in Germany? What is the RMS voltage?

Task 2: Sector Coupling

a) Explain the concept of sector coupling.

b) What are the advantages of sector coupling?

c) What are the disadvantages of sector coupling?

d) In the lecture we discussed real and virtual energy conversion. Based on an example explain the difference between real and virtual energy conversion.

Task 3: Assumptions and Implications of the Power Grid

a) Can the frequency be used to check the balance between demand and supply? How?

b) State and explain the traditional assumptions of energy management and control.

c) The introduction of renewable energy sources brought new implictations to the assumptions of energy management and control. Which challenges did these implications introduce to the energy management?